

JK 12-16 Python First Programmer
Lesson 07 : Arrays

Duration : 90 mins

Description

Students learn what is an array and Implement them in their projects. In this lesson students create a few projects to practice using arrays.

Objectives

Students will be able to:

1. Understand what is arrays and use it in a code.
2. Implement arrays in the project.
3. Analyze code and figure out the outcome of the code.
4. Create projects combining all concepts learned during the lesson.

Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
5 mins	Comprehension Check
10 mins	Arrays Introduction
10 mins	Arrays Introduction (Continued)
10 mins	Length of Array
15 mins	Adding and Deleting Array Element
10 mins	Loop Through Array
20 mins	Mini Projects
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students.	
Comprehension Check (5 mins)	
- Teacher ask students if they are still remember about the materials from the previous lesson. - Teacher check on students' comprehension regarding the previous lesson by asking relatable questions. - Teacher explain and demonstrate in brief or summarize about what they have learned from the previous lesson.	
Arrays Introduction (10 mins)	
Explain – The Concept of Arrays - Before we start, we first show students the “Array explained in 1 minute” video and ask students what they think arrays are? - We then follow up and explain that arrays are just like lists that we can use to store the information that we would like to store.	

- But just like the video has explained, there are certain limitations when using an array. This limitation is that when using arrays, we have to remember that the type should be the same.

Discussions – Why Arrays?

- We continue and ask students why we would like to use arrays? But before we start delving into the usage of arrays / lists in programming, we should first think about why we need lists in our daily lives?
- We tell students to pair up and discuss between each other on what lists can be used for in our daily lives. We give students some time and ask them about their answers.
- We tell students that there are a lot of different ways we can use arrays / lists in our daily lives. For example, lists / arrays can help us:
 - Store similar data in one place.
 - Organize informations.
 - Analyze data that are related to each other.
 - Think about the big picture.
- And so, just like in real life, having a list / arrays in programming could also help us in the same way.

Arrays Introduction Continued (10 mins)

Explain and Show – My First Array

- We show students the code and explain to students that when we want to declare an array, the steps are quite similar compared to when we would like to declare a variable. The difference is that when declaring an array, the content of it should be located inside the square bracket ([]), and each element should be separated by a coma.
- To print the content of a specific array index, we can call the array name followed by a square bracket and the index. (Example: MyArray[2]).

File/Code:

[1.MyFirstArray.py](#)

- Next, we ask students about what they think will happen when we run the project. What do they think will be shown?
- After students have answered, we then show the project and ask students if their guess was correct and explain that in programming, an array index will always start with 0. Therefore, if we write for example, `MyArray[1]`, what will be shown is the second data that we have inserted.

Individual Activity – Creating Arrays

- We then tell students to try and create their own arrays by following the steps that have been previously explained by the teacher.
- We tell students that they should at least create two arrays that store two different data types (Strings and Integer)

Length of Array (10 mins)

Explain and Show – Concatenation

- We explain that in order to concatenate between our arrays, we can do so just like how we concatenate variables. But as a note, we remind students that if the content of our arrays are integers, we need to convert the data type first.

Explain and Show – Array Length

- We continue on and explain that there is also a function which we can use to check the length of an array. This can be done by using the `len(nameofarray)` function.
- We explain that later on this can be useful if we want to loop through an array.

Individual Activity – Adding Concatenation and Array Length

File/Code:

[2.LengthOfArray.py](#)

- We tell students to add the components that have been explained to their project. We give them time to do the project and check their results when they are done.

Adding and Deleteng Array Element (15 mins)

Explain and Show – Array Addition

- We show students the code and explain that to add a number into the array that we have created, we can use the `.append("contenttoadd")` function.

Explain and Show – Array Deletion

- We explain that to delete an array, there are two ways of doing it. The first one is to use the `del` function and the second one, we can use the `array.remove("Content")` function.
- We show students the code and explain to students how both of the functions work. We explain that when we use the `del` function, we have to **specify the array and array index**.
- If we use the `.remove()` function, we can **specify the content of the array**.

Individual Activity – Program with Array Addition and Deletion

- We tell students that they should create a program which incorporates both of the functions that we have previously discussed.
- We tell them that they can try creating a more advanced version if they want to try out what they have learned so far. (for example, we can tell students that they can try to create a program which will ask if they would like to add or remove an array).

File/Code:

[3.AddAndDelete.py](#)

Loop Through Array (10 mins)

Explain and Show – Loops & Arrays

File/Code:

[4.LoopThroughArray.py](#)

- We explain to students that later on, there will be a lot of situations where we will need to go through all of the content of our arrays and check the contents of it.
- Therefore, it is really important for us to be able to loop through our arrays. To do this, there are two ways that we will learn today.
- The first way is to use a while loop. As have been displayed in the project, here, we will loop through the array by using the l variable as our indicator and making the length of our array as our max value.
- The second way is to use the for loop. This function can be used to loop through all of the content of our array and insert it to a variable called color each time we loop through it. This variable can then be used just like how we would use normal variables.

Individual Activity – Adding Loops and Arrays

- We let students create a new project and tell them to add at least 1 while loop and 1 for loop. We tell students to explore further by themselves in which afterwards, we can let them know that they could try helping other students if they are done.
- We encourage students to try the projects created by other students.

Mini Projects (20 mins)

Explain – Mini Projects

- We show students the project without showing the code and demonstrate what can happen in the program.

Group Activity – Flowchat / Pseudo-Code Creation

- Now that they have a general idea on how the project might work, we ask students to pair up and create the pseudo code / flowchart of the program so that later on it will be much easier for them to create the program.

File/Code:

[5.IntOrString.py](#)

[6.MiniProject.py](#)

- We give students time to do it by themselves and tell them that if they want us to run the project again, they can raise their hands.
- When they are done, a person from each group should come forward and write their flowchart / pseudo-code on the white board.

Explain and Code – Integer or String Checker

- Next, we explain that in order to create the project, there is one new concept that they should understand. In this case is a way so that the program could differentiate between an integer and a string.
- We open up the IntorString.py project and explain to students that to check if a variable is a string, they could use the `.isalpha()` function which can be seen in the project code. And to check if it is a digit, we can use the `.isdigit()` function.
- We give students some time to follow us as we demonstrate the project.

Individual Activity – Mini Project Creation

- Lastly, we tell students that they should try and create the project individually first.
- We check the student's progress and when the time is almost up, we tell them that we will discuss the project together.

Discuss – Mini Project

- We discuss the project with the students by asking them if they were able to finish their project, if they could, what helped them create their project. And if not, did they encounter a problem?
- We show students the code and go through the project together. We can also ask those who might not have finished to see if there are parts of their code that might be missing.

Wrap Up (5 mins)

Wrapping Up

- We sum up by asking questions, making sure the students remember the most important things.
- We thank students for doing a great job overall.

☐ ★ Good Luck With The Class !! ★ ☐